Numerical Optimization 13: Sampling Plans

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May 17, 2020

Overview

1. Sampling
2. Full Factorial
3. Random Sampling
4. Uniform Projection Plans
5. Stratified Sampling
6. Space-Filling Metrics
7. Summary

Optimization with expensive function evaluations

For many optimization problems, function evaluations can be quite expensive.

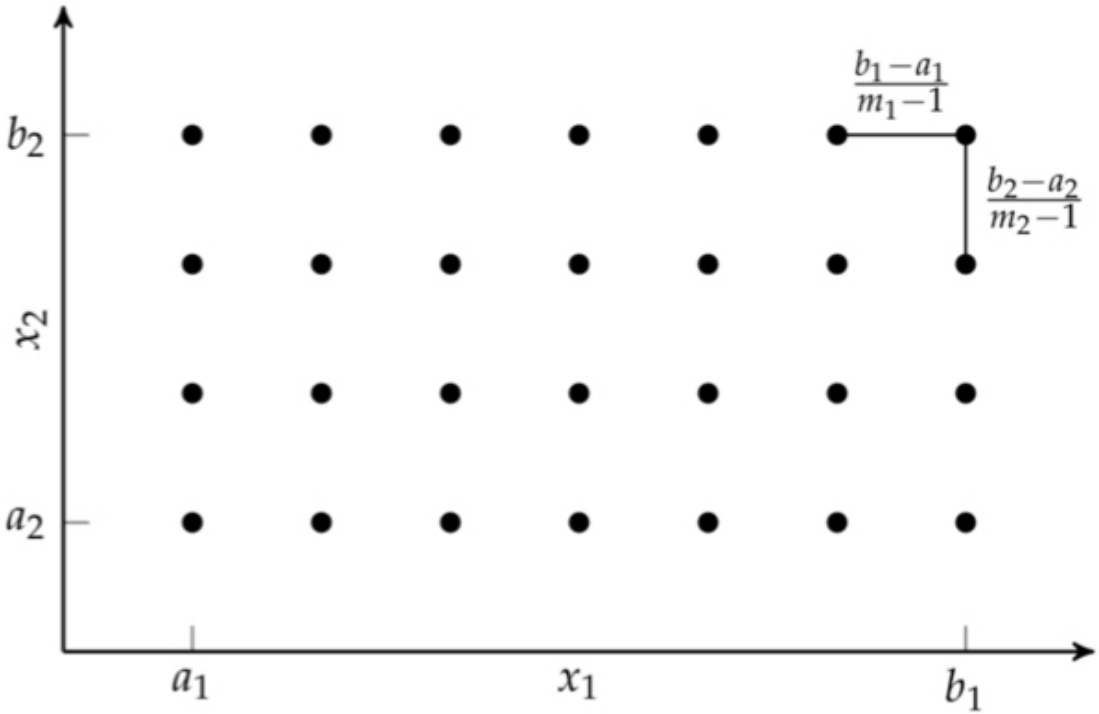
- an aircraft design may require a wind tunnel test
- deep learning hyperparameters may require a week of GPU training
- ...

A common approach for optimizing in these contexts is to build a surrogate model, Further evaluations of the true objective function can be used to improve the model. Fitting such models requires an initial set of points, ideally points that are space-filling; that is, points that cover the region as well as possible.

Full Factorial

The full factorial sampling plan places a grid of evenly spaced points over the search space.

- a lower/upper-bound vector a, b such that $a_i \leq x_i \leq b_i$
- m_i samples in each x_i separated by a distance $(b_i - a_i)/(m_i - 1)$



Random Sampling

In some cases, it may be possible to transform a problem so that constraints can be removed. For example, bound constraints a x b can be removed by passing x through a transform

Uniform Projection Plans

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Summary

- Constraints are requirements on the design points that a solution must satisfy.
- Some constraints can be transformed or substituted into the problem to result in an unconstrained optimization problem.
- Analytical methods using Lagrange multipliers yield the generalized Lagrangian and the necessary conditions for optimality under constraints.
- A constrained optimization problem has a dual problem formulation that is easier to solve and whose solution is a lower bound of the solution to the original problem.